CCC-W-51D
June 6, 1975
SUPERSEDING
CCC-W-0051C(DSA-DM)
September 12, 1967 and
CCC-W-51B
May 7, 1958

FEDERAL SPECIFICATION

WADDING, COTTON, SURGICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers surgical cotton wadding for use under Plaster of Paris bandages and casts.

1.2 Classification.

1.2.1 Type and size. Cotton wadding shall be of one type, 5 inches wide by 6 yards long.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

FSC 6510

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Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap: Water-Vaporproof or Water proof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of inviation for bids or request for proposal shall apply.

U.S. Pharmacopeial Convention Publication:

Pharmacopeia of the United States

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

American Society for Testing and Materials (ASTM) Standard:

D 828-60 Tensile Breaking Strength for Paper and. Paper Products.

(Application for copies should be addressed to American Society for Testing and Materials, 1961 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be a good grade of nonabsorbent carded cotton, 35 percent of the fibers to be not less than 1/2 inch in length. Fifty percent of the fibers used may be bleached.

3.2 Tolerance. A tolerance in width of plus or minus 1/4 inch will be permitted. A tolerance in length of plus or minus 3 inches will be permitted.

3.3 Absorbency. When tested in accordance with paragraph 4.4.3, the test pieces shall not submerge within 15 minutes.

3.4 Color. Color shall be white, and wadding shall contain no soluble coloring matter. (See 4.4.4.)

3.5 Sizing. Both sides of layers shall have a glaze coating of adequate thickness and bursting strength to resist tearing and to give a smooth finish.

3.6 Weight. Weight shall average 1.6 ounces per square yard with no individual piece weighing less than 1.3 ounces per square yard (see 4.4.5).

3.7 Formation of roll. Strip of one continuous piece shall be tightly rolled in bandage form.

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3.8 Tensile strength. Tensile strength shall be two pounds, plus or minus one pound, for one ply of wadding cut one inch wide by six inches long, cut in the lengthswise direction. (See 4.4.6.)

3.9 Workmanship. Workmanship shall be first class throughout. The cotton wadding shall be clean and free from oil stains, metallic particles and other foreign matter. Loose ends, ravelings, or other defects which detract from the appearnace or impair the serviceability shall not be permitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the wadding, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For visual examination. Sampling for visual examination shall be conducted in accordance with MIL-STD-105. Sample unit shall be one roll.

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4.2.2 For tests. Sample size for tests shall be in accordance with inspection level S-2 of MIL-STD-105, except that the acceptance number shall be zero for all sample sizes. A minimum of three samples shall be taken, except as specified herein. (See 4.4).

4.3 Examination. The cotton shall be examined for conformance with requirements for size, color, sizing, continuity and workmanship. The classification of defects shall not be limited to those listed in Table I.

Table I. Classification of Defects

Defect		Classification Major	Minor
101	Width or length not as specified.	X	
102	Dirt, stains, and foreign matter.	X	
103	Not glazed.	X	
104	Color other than white.	X	
201	Presence of loose ends or ravelings.		X
202	Not one continuous piece.		X
203	Not uniform thickness or not tightly or evenly rolled.		X

4.3.1 Acceptable quality levels and inspection levels. The inspection levels and acceptable quality levels, expressed in defects per hundred units (DHU) shall be as shown in table II.

Table II. Acceptable quality levels

Defect	AQL	Inspection level
Major	1.0	II
Minor	2.5	II

4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

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Table III Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packing and Packaging Material	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size location, sequence, or method of application.

4.4 Tests. Tests shall be conducted on samples selected in accordance with 4.2.2 as follows.

4.4.1 Conditioning. Unless otherwise specified, samples for testing shall be conditioned a minimum of four hours at 21° ± 1.1° Centrigrade (C.) (70° ± 2° Fahrenheit (F.)) and 65 ± 2 percent relative humidity, before test performance.

4.4.2 Fiber length. Test for fiber length shall be made as specified in the latest revision of the United States Pharmacopeia.

4.4.3 Absorbency. A ten gram portion of cotton folded into a square approximately four inches on a side, when placed on the surface of a suitable volume of distilled water in a container of suitable size such as a one liter-capacity, Griffin type, beaker, at room temperature, shall meet the requirements set forth in paragraph 3.3.

4.4.4 Test for uncombined dyes (no soluble coloring matter). Pack ten-grams of absorbent gauze in a narrow percolator, and extract slowly with ethyl alcohol until the percolate measures 50 milliliters (ml.). When observed downard in a column 20 centimeters (cm.) in depth, a yellowish color, is permitted.

4.4.5 Weight. Test specimens shall be taken from not less than ten rolls to determine compliance with paragraph 3.6.

4.4.6 Tensile strength. Test for tensile strength shall be made in accordance with ASTM Standard D828-60, conditioning of sample shall be as specified by the ASTM method. Sample size of wadding shall be one inch by nine ply, cut six inches long from the length. Jaw separation shall be five inches. The force required to break the sample shall be divided by nine to determine the mean tensile strength of a single ply. (See 3.8).

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One package containing twelve (12) rolls of wadding as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package. Each roll shall be tightly rolled and secured in a paper wrapper. Twelve (12) rolls shall be further packaged in a bag fabricated from virgin polyethylene seamless tubing having an average wall thickness of 0.004 inch. Excess air shall be expelled or evacuated from the bag and closure effected by heat sealing.

5.1.1.1.2 Intermediate package. Twelve (12) units of issue shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing, Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty four (24) units of issue shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with PPP-B-636.

5.2.2 Level B. Twenty four (24) units of issue shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED Std No. 123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name and address of manufacturer shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cotton wadding is to be used under plaster of paris bandages and casts, to prevent soreness and chafing.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6510-00-204-2000	WADDING, COTTON, SURGICAL SHEET, 5 inches by 6 yards, 12s.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No: 6510-0610

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Prices 30 cents each.

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OMB Approval
No. 22-R255

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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